

$$\mathcal{S}_{\text{total}} = \int_{\mathcal{M}} d^4x \sqrt{-g} \, \mathcal{L}_{\text{univ}} + \mathcal{S}_{\partial\mathcal{M}}$$

$$\begin{aligned} \mathcal{L}_{\text{univ}} = & \frac{c^4}{16\pi G} e^{-2\phi} (R - 2\Lambda_0) - \frac{1}{2} g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi - \frac{1}{2} m_\phi^2 \phi^2 - \frac{\lambda_\phi}{4!} \phi^4 \\ & - \frac{c^4 a_0^2}{8\pi G} \left[\sqrt{\frac{g^{\mu\nu} \partial_\mu \chi \partial_\nu \chi}{a_0^2} \left(1 + \frac{g^{\alpha\beta} \partial_\alpha \chi \partial_\beta \chi}{a_0^2} \right)} - \text{asinh} \left(\sqrt{\frac{g^{\mu\nu} \partial_\mu \chi \partial_\nu \chi}{a_0^2}} \right) \right] \\ & + \frac{1}{2} g^{\mu\nu} \text{Tr} \left((\nabla_\mu V - i A_\mu V)^\dagger (\nabla_\nu V - i A_\nu V) \right) - \frac{\gamma_0}{4} \left(\frac{1-\chi}{1+\chi} \right) g^{\mu\alpha} g^{\nu\beta} \text{Tr} (\mathcal{F}_{\mu\nu}[V, A] \mathcal{F}_{\alpha\beta}[V, A]) \\ & - \frac{1}{4g^2(\mu)} g^{\mu\alpha} g^{\nu\beta} \text{Tr} \left(F_{\mu\nu}^{E_8} F_{\alpha\beta}^{E_8} \right) + \frac{\theta_{\text{QCD}}}{32\pi^2} \frac{\epsilon^{\mu\nu\alpha\beta}}{2\sqrt{-g}} \text{Tr} \left(F_{\mu\nu}^{E_8} F_{\alpha\beta}^{E_8} \right) \\ & + \sum_{g=1}^3 \bar{\psi}_g \left[i e_a^\mu \gamma^a \left(\partial_\mu + \frac{1}{4} \omega_\mu^{bc} \sigma_{bc} - i g A_\mu \right) - M_{\text{Pl}} e^\phi e^{-\lambda_g/\kappa_Y} (1 + \xi_g (\chi - \chi_Y)) \right] \psi_g \\ & - \sum_{g,g'=1}^3 Y_{gg'} \phi \bar{\psi}_g \psi_{g'} \end{aligned}$$

$$\mathcal{S}_{\partial\mathcal{M}} = \frac{c^4}{8\pi G} \oint_{\partial\mathcal{M}} d^3x \sqrt{|\gamma|} e^{-2\phi} K + \frac{k_B c^3}{4G\hbar} \oint_{\mathcal{H}} dA \left(1 - \frac{I(A:B)}{I_{\text{max}}} \right)$$

$$\mathcal{F}_{\mu\nu}[V,A] = [V^\dagger(\nabla_\mu V - iA_\mu V),\, V^\dagger(\nabla_\nu V - iA_\nu V)], \quad V \in V_2(\mathbb{R}^3) \cong SO(3)$$

$$\text{Out}(\text{Spin}(8)) \cong S_3 \implies \mathbb{Z}_3, \quad g \in \{1,2,3\}$$

$$a_0 = \frac{cH_0}{2\pi}, \quad \Sigma_c = \frac{a_0}{2\pi G}, \quad \ell_P = \sqrt{\frac{\hbar G}{c^3}}$$

$$\chi_Y = \frac{1}{\sqrt{2}}, \quad \kappa_Y = \sqrt{\theta(2-\theta)} = 0.953939 \dots \quad (\theta = 0.7)$$

$$g_{\text{bare}}(M_{\text{Pl}}) = \frac{\kappa_Y \chi_Y}{\pi} \approx 0.2147, \quad \tilde{g}_{\mu\nu} = g_{\mu\nu} + 2\ell_P^2 \nabla_\mu \chi \nabla_\nu \chi \implies \left| \frac{v_{\text{gw}}}{c} - 1 \right| = 0$$